

MANISH PATIYAL

Senior Principal Technical Lead | Mobile & Generative AI Solution Architect

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PROFESSIONAL SUMMARY

Senior Principal Technical Lead with 12+ years of experience designing and delivering scalable Android, React Native, and Flutter applications for global clients. Strong background in solution architecture, native SDK integration, technical leadership, pre-sales consulting, and client engagement. Currently expanding into Generative AI solution architecture — building working knowledge of Python, LLM application design, RAG pipelines, and LangChain — with the goal of bridging enterprise mobile architecture experience with GenAI-powered product design.

SKILLS

Languages: Python, Kotlin, Java, JavaScript, TypeScript, Dart, Kotlin Multiplatform (KMP)

Mobile: Android, React Native, Flutter, Cross-Platform Mobile Development

Android: Jetpack Compose, Coroutines, Flow, Room, WorkManager, Hilt, Navigation, MVVM, Clean Architecture, Retrofit, Firebase

React Native: Turbo Modules, Native Bridge Development, Third-Party SDK Integration, REST APIs

Generative AI / LLM Frameworks (in progress): LangChain, LangGraph, LlamaIndex, RAG, AI Agents, Tool Calling, MCP (Model Context Protocol)

LLM & Cloud AI Platforms (in progress): OpenAI, Anthropic Claude, AWS Bedrock

Vector Databases (in progress): FAISS, Pinecone, Qdrant

AI Engineering & Backend (in progress): Python, FastAPI, Prompt Engineering & Guardrails

AI Quality & Ops (in progress): MLOps (MLflow, CI/CD), AI Evaluation (RAGAS, DeepEval), AI Observability

AI Security & Architecture (in progress): AI Security (prompt injection defense, data protection), Enterprise AI Architecture

ML Fundamentals (in progress): scikit-learn, pandas, NumPy

On-Device ML (in progress): TensorFlow Lite, Core ML, ONNX

Dev Tools & Cloud: Android Studio, Xcode, Cursor, Docker, Kubernetes, AWS Serverless (Lambda, API Gateway, Step Functions, IAM), Firebase, GitHub Actions, Bitrise, Jenkins

Version Control / Tracking: Git, GitHub, GitLab, Jira, Mantis

Languages Spoken: English, Hindi

EXPERIENCE

VT Netzwelt Pvt. Ltd., Chandigarh

May 2017 – Present

Senior Principal Technical Lead | Jan 2022 – Present

- Own end-to-end technical solution architecture for global Real Estate, Retail, and Consumer Research clients — leading a cross-functional team delivering React Native, Flutter, and Android applications.
- Serve as primary technical point of contact for stakeholders across Europe and North America, shaping product roadmaps and resolving critical delivery and architecture challenges.
- Drive pre-sales strategy — technical discovery, solution design, effort estimation, and client proposals/presentations — directly influencing new and expanded client engagements.
- Raise engineering quality through architecture reviews, RCA/CAPA processes, and sprint retrospectives, while mentoring developers through the full implementation lifecycle.
- Built a custom Insta360 native bridge (Android/iOS) powering immersive 3D property visualization for a flagship real estate client.

Senior Software Engineer | May 2017 – Jan 2022

- Built the GfK Consumer Scan app from the ground up in React Native — scaled to 5M+ downloads on Google Play across Android and iOS.
- Delivered core production features — barcode scanning, offline sync, camera integration, and REST API communication — for a global consumer research client.

- Partnered with PMs, designers, QA, and international clients to turn requirements into technical specs and high-quality releases across the App Store and Google Play.
- Mentored engineers through code reviews and internal technical seminars on Android and React Native best practices.

Software Engineer — *IDEMIA, Noida*

Aug 2013 – May 2017

- Delivered Android-based biometric solutions — enrollment, authentication, and identity verification — for Healthcare, Telecom, and Security clients using proprietary SDKs.
- Engineered a Remote SIM Personalization application enabling secure SIM provisioning on IDEMIA tablets, eliminating the need for traditional SIM programming infrastructure.
- Partnered with Reliance Jio's technical team to integrate biometric authentication into customer and agent onboarding workflows at scale.
- Supported the full product lifecycle — solution design, SDK integration, technical demos/POCs, and deployment — for enterprise clients across India and international markets.

GENERATIVE AI LEARNING & CONTINUING EDUCATION

- Currently building applied Generative AI skills: LLM application design, prompt engineering, RAG pipelines, LangChain, and vector databases, alongside ML fundamentals (scikit-learn, pandas, NumPy) and on-device ML (TensorFlow Lite, Core ML, ONNX).
- Enrolled in “Complete Generative AI Course With LangChain and Huggingface” (Krish Naik) — covering Generative AI fundamentals, LangChain, Huggingface model integration and fine-tuning, RAG pipelines, and building/deploying real-world applications such as chatbots and content generators to cloud and on-premise environments.
- Goal: apply 12+ years of mobile solution architecture and client-facing technical leadership experience to GenAI/LLM-powered product architecture.

EDUCATION

Lovely Professional University, Jalandhar — B.Tech (CSE)

June 2010 – Aug 2013 | Relevant coursework: Artificial Intelligence, Data Structures, Operating Systems, Computer Networks, DBMS.